	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 6

Lab Title: **Design and Configuration of Half and Full Subtractor**

Student Name: **Muammad Waqas** **Reg. No:** **240756**

Objective: To experimentally verify the operation of basic Subtractor,various configurations of subtractor.

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks: _____

Date: _____ Signature: _____

TASK1:

Design a Half subtractor circuit and complete the truth table.

Steps:

1. Complete the truth table.
2. Write the logic equations for Difference and Carry.
3. Draw the complete logic diagram for the Half Subtractor.
4. Verify your results practically.

A	B	D	Bo
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

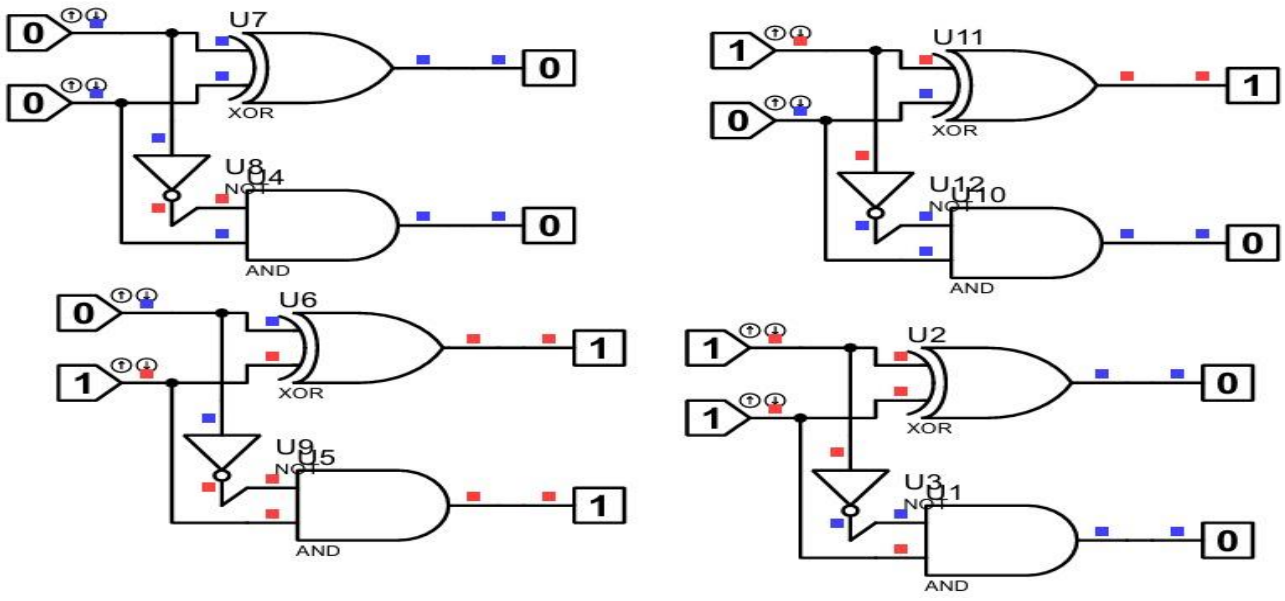
2 Logic Equation for Difference and borrow

Difference	$A \oplus B$
Borrow	A'.B

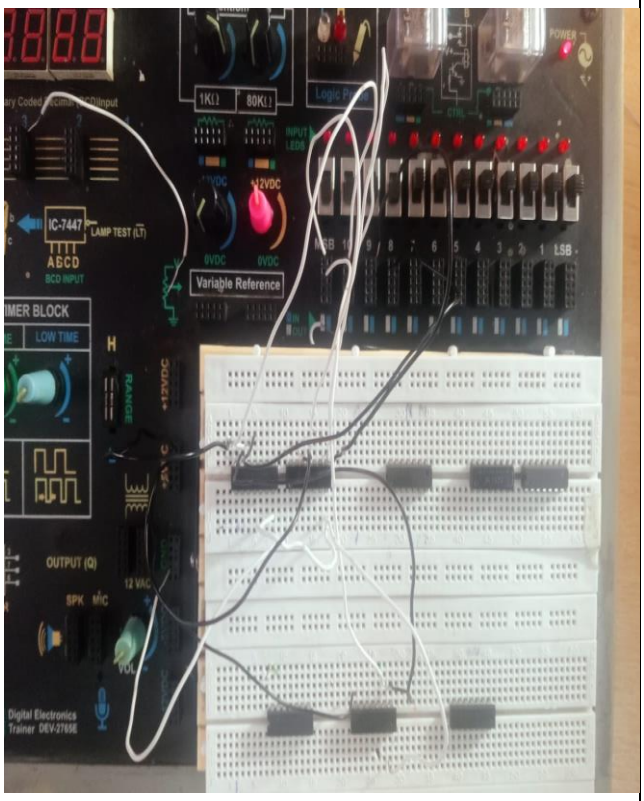
Difference= $A \oplus B$

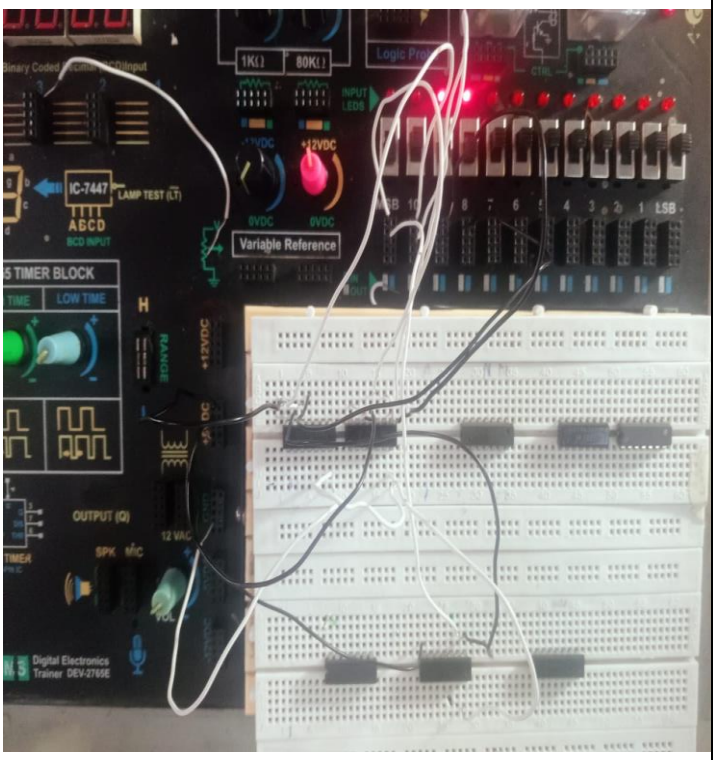
borRow= $A \cdot B$

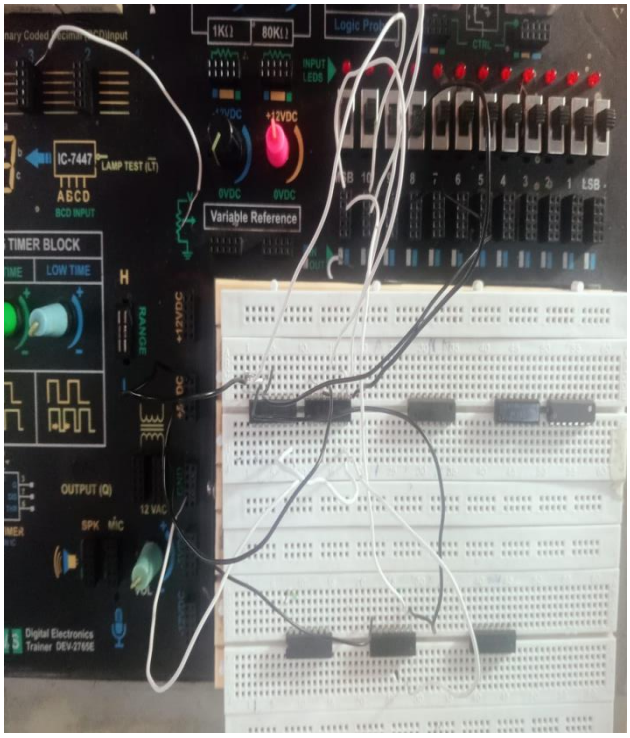
3. Complete diagram of Half Subtractor:

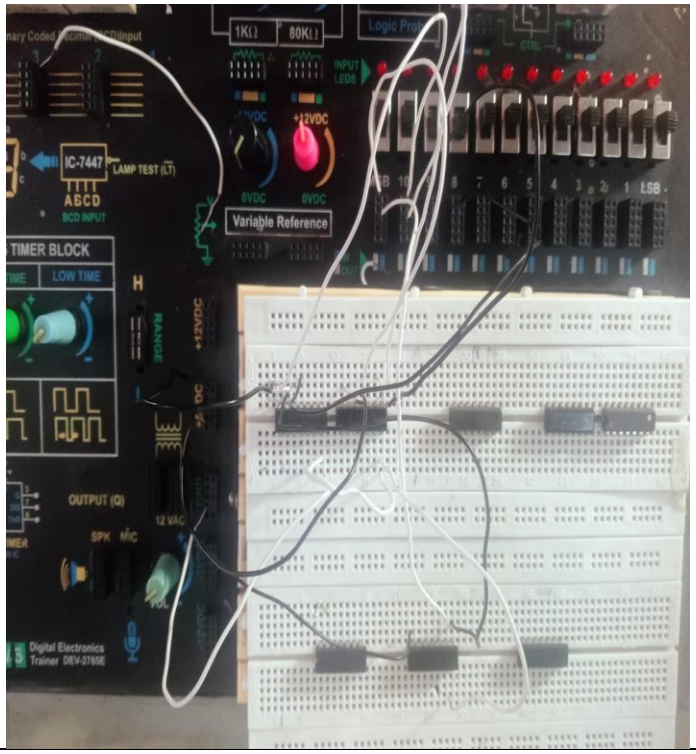


4. Practical simulation:







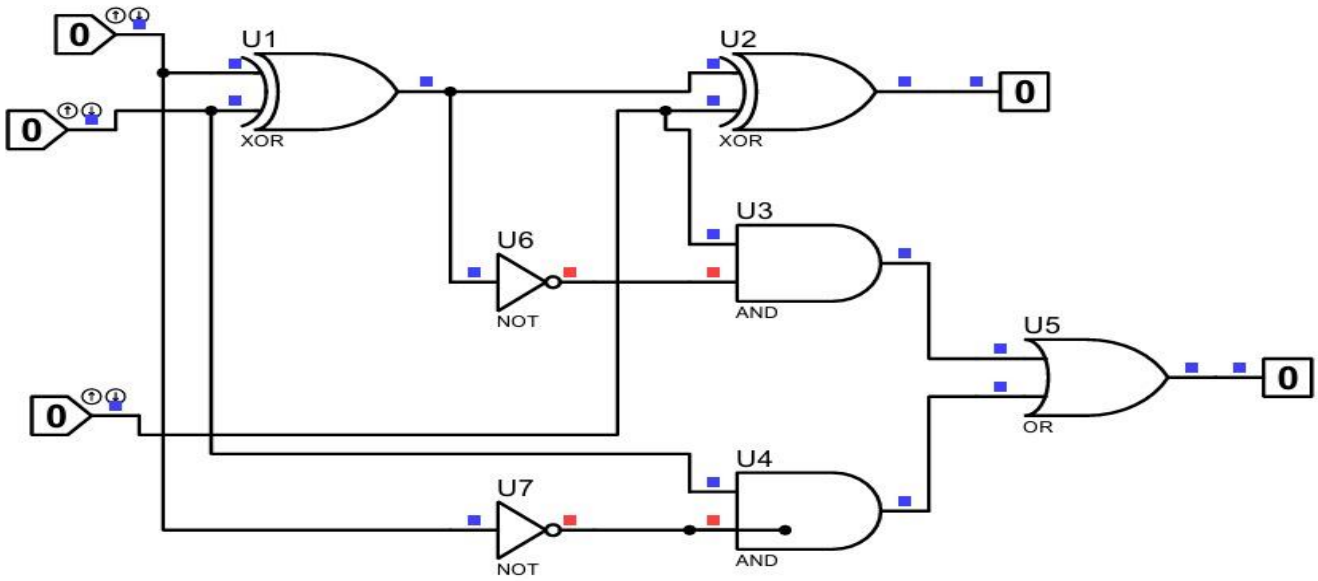


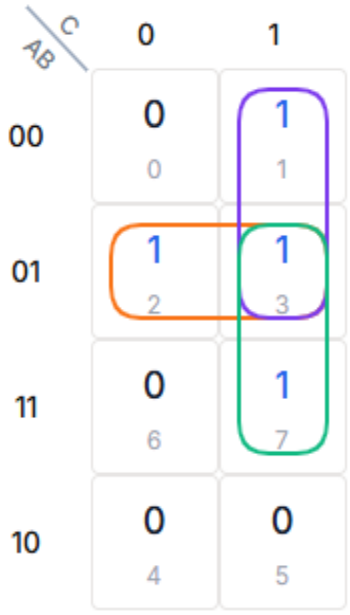
Task 2:

Design a Full Subtractor circuit and complete the truth table.

Steps:

1. Draw the complete logic diagram.
2. Write the logic equations for Difference and Borrow.
3. Complete the Truth Table.
4. Implement the circuit on trainer and verify your results practically.



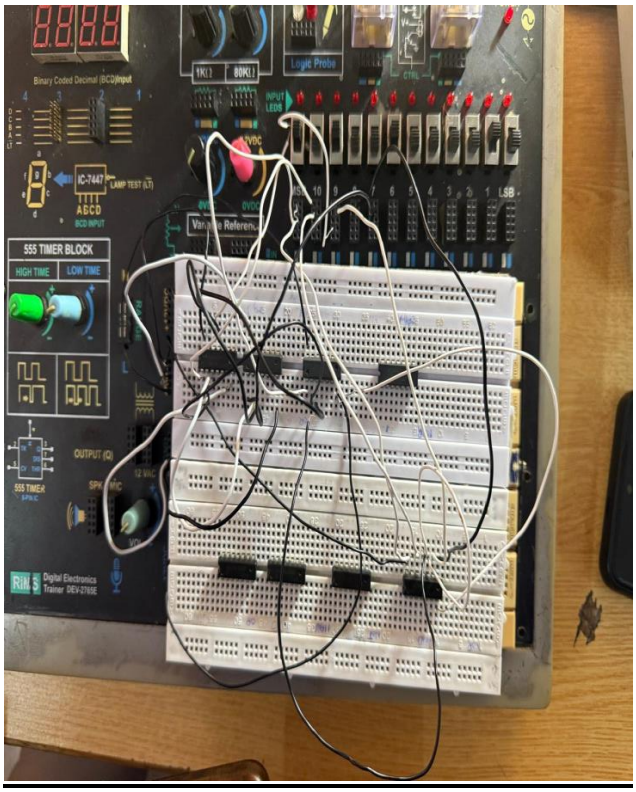
K-Map for difference	K-map for Borrow
	
Simplified expression: A'B'C+A'BC'+AB'C'+ABC	**Simplified expression:** A'B+A'C+BC

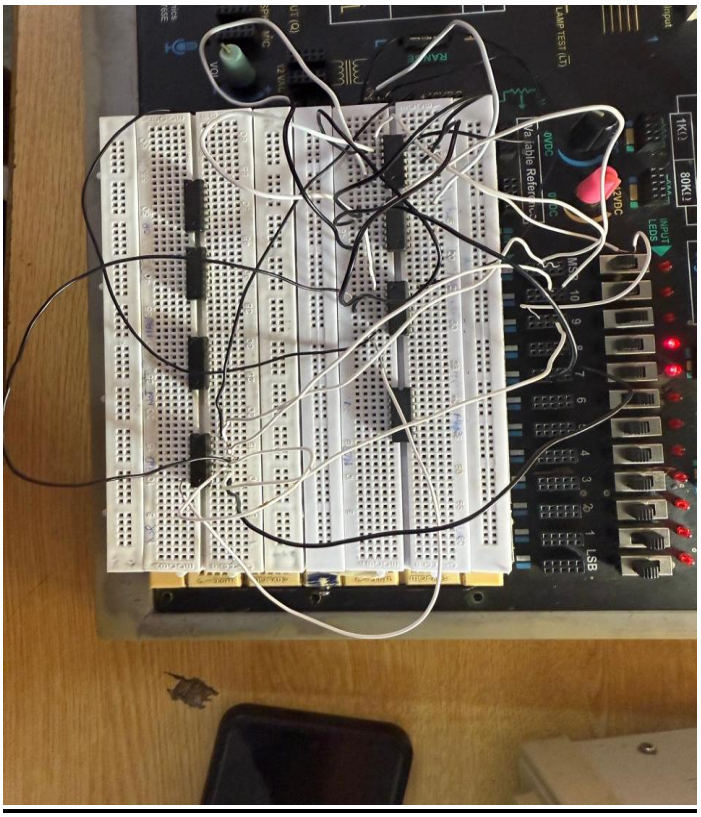
Difference	Borrow
1. Difference (D) $D = b_i'(A'B + AB') + b_i(A'B' + AB)$ $D = b_i'(A \oplus B) + b_i(A \oplus B)'$ $D = A \oplus B \oplus b_i$	**2. Borrow-out (b)** $b = A'B'b_i + A'B\, b_i' + A'B\, b_i + A\, B'b_i$ $b = A'B(b_i' + b_i) + A'B'b_i + A\, B'b_i$ $b = A'B + A'B'b_i + A\, B'b_i$ $b = A'B + b_i(A'B' + A\, B')$ $b = A'B + (A \oplus B)'b_i$

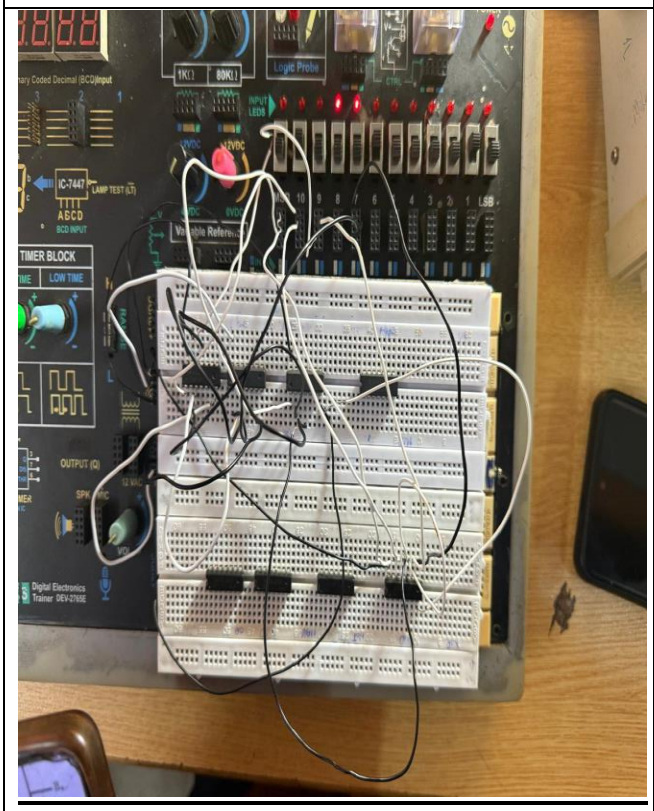
TRUTH TABLE:

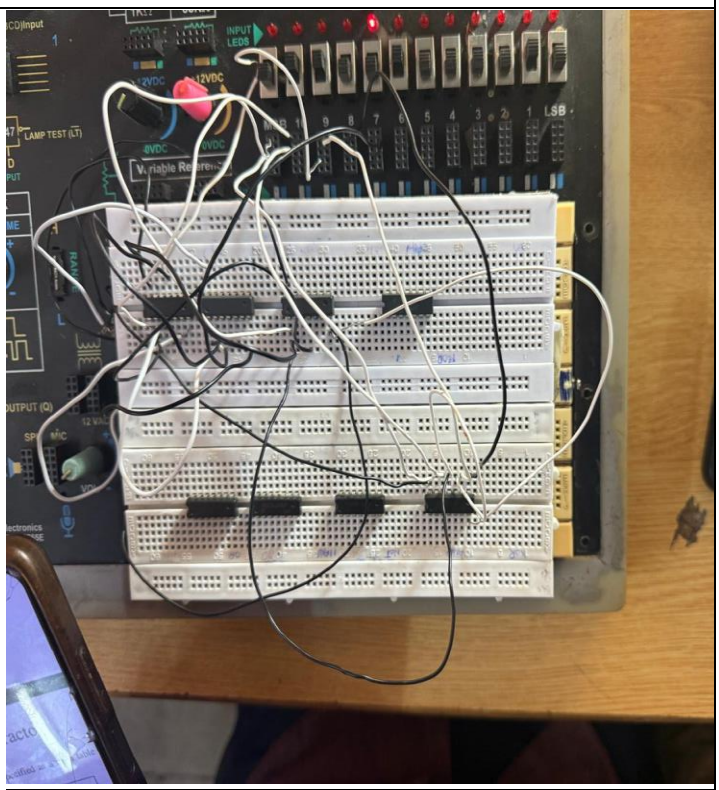
ROW	A	B	Bin	D	Bout
m0	0	0	0	0	0
m1	0	0	1	1	1
m2	0	1	0	1	1
m3	0	1	1	0	1
m4	1	0	0	1	0
m5	1	0	1	0	0
m6	1	1	0	0	0
m7	1	1	1	1	1

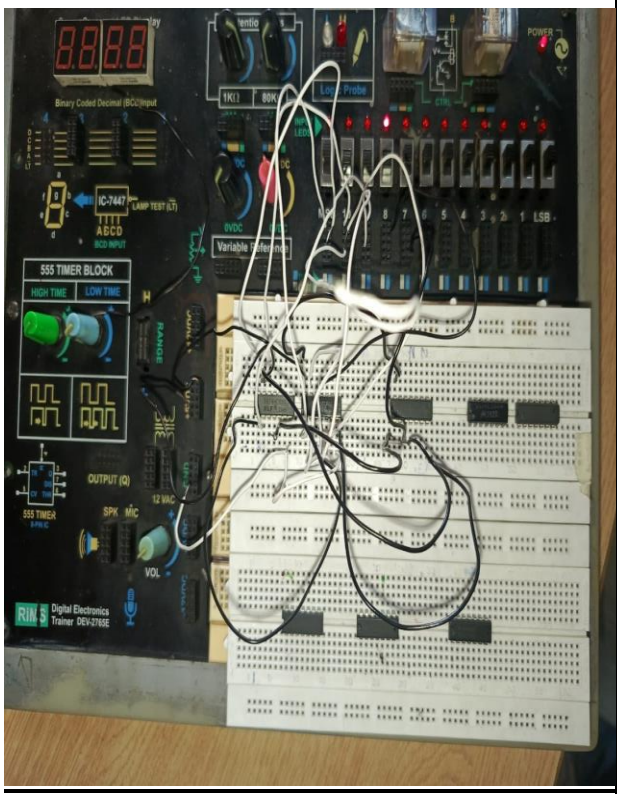
Practical Simulation:

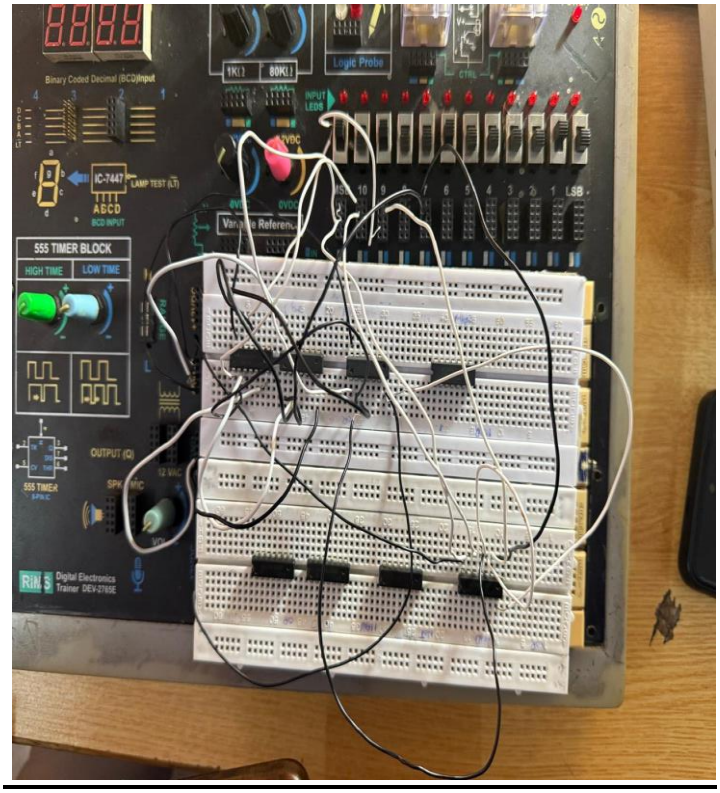


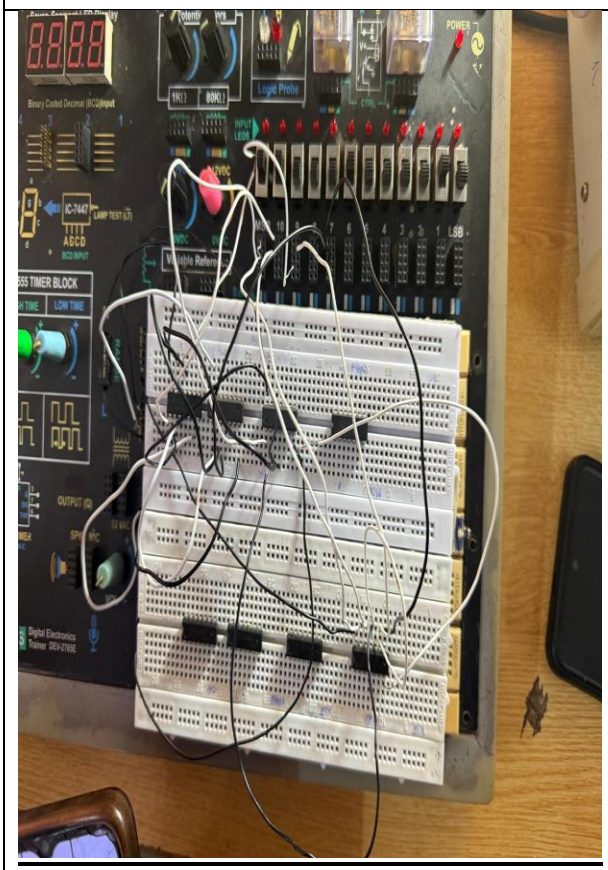


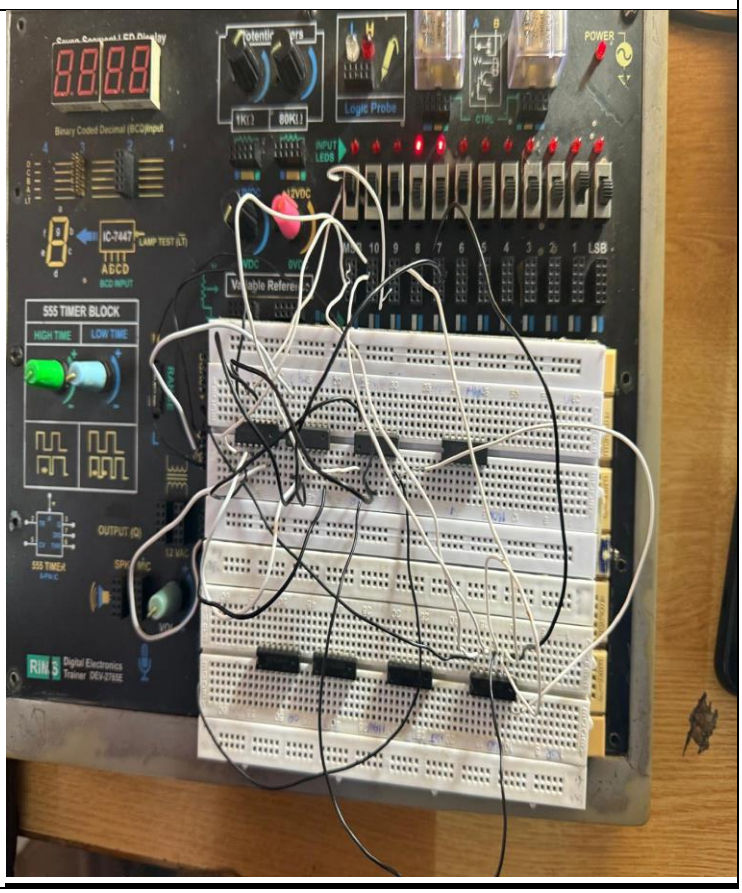





555 TIMER BLOCK
HIGH TIME
LOW TIME
OUTPUT (Q)
555 TIMER
SPK
MIC
VOL
Digital Electronics
Trainer DEV-2705E


555 TIMER BLOCK
HIGH TIME
LOW TIME
OUTPUT (Q)
555 TIMER
Digital Electronics
Trainer DEV-2705E


OUTPUT (Q)
Digital Electronics
Trainer DEV-2705E


555 TIMER BLOCK
HIGH TIME
LOW TIME
OUTPUT (Q)
555 TIMER
Digital Electronics
Trainer DEV-2705E

Conclusion:

In digital logic design, a subtractor is a fundamental combinational circuit used to perform binary subtraction. It takes binary inputs and generates two outputs: the difference and the borrow. The design and configuration of subtractors are similar to adders, but instead of adding binary numbers, they perform subtraction. The Half Subtractor has two inputs, A and B, and provides two outputs — the Difference (D) and the Borrow (b). It performs subtraction of two single-bit binary numbers, where the difference is given by A $\oplus$ B and the borrow by A'B. However, it cannot handle a borrow from a previous stage.The Full Subtractor can perform multi-bit subtraction by connecting several subtractor circuits in sequence. In conclusion, subtractors are essential components in digital arithmetic systems and play a key role in devices such as arithmetic logic units (ALUs) and processors, where binary subtraction operations are required.